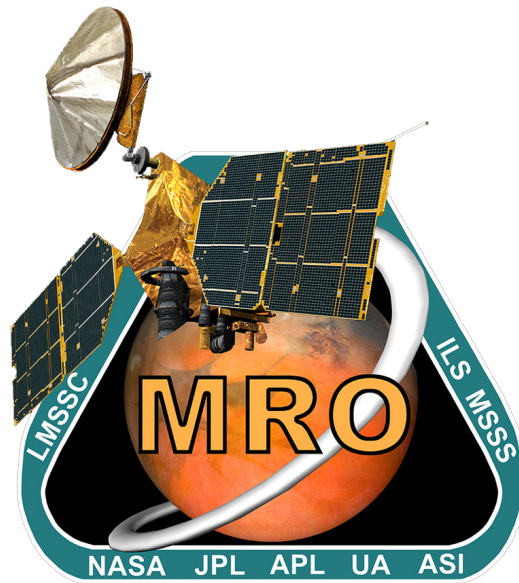




MRO Reverse Engineering Study

Assignment #6

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1 Electric Power Subsystem

The Electric Power Subsystem (EPS) generates, stores, distributes and controls electrical power within Mars Reconnaissance Orbiter (MRO). The EPS is designed around a central bus, which operates at a voltage of 28 ± 4 V [9]. This subsystem supplies a continuous source of power to the various electrical loads of the spacecraft during all phases and modes. The following paragraphs aim at providing a complete overview of the EPS, outlining its main features and explaining the rationale behind the solutions adopted.

1.1 Architecture and rationale

The power is generated using two solar arrays and two nickel-hydrogen batteries. The solar panels are mounted on opposite sides of the orbiter with a gimbaled connection, allowing continuous Sun tracking with low aspect angles. The battery supplies the required power during the modes in which solar power generation is not possible, even in case of failure, thanks to the second redundant battery. Figure 1 provides a comprehensive overview of the overall logic and functional architecture of the EPS. It illustrates the main components involved, their interconnections, and the flow of electrical power from generation to distribution and storage. Each of these elements will be analyzed in detail in the following sections, highlighting both their functional roles and design considerations.

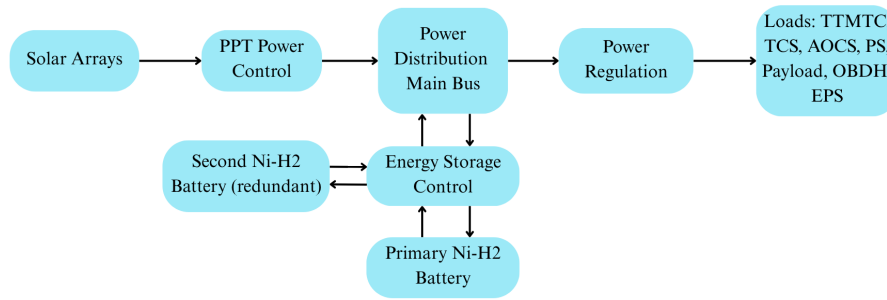


Figure 1: EPS block scheme

1.1.1 Solar Arrays

Solar panels are lighter and less complex than other energy sources, such as radioisotope thermoelectric generators (RTGs), making solar arrays more suitable for missions wherein spacecraft mass is a critical factor. Thus, as its primary means of power generation, MRO employs two solar arrays placed at either side of the main body. These solar arrays are in a folded configuration at launch and are deployed during checkout (as shown in Figure 2) and remain deployed throughout the entire mission. The selection of solar arrays is also driven by aerobraking maneuver, providing a large area for drag interaction with the Martian atmosphere, saving propellant mass.

Each array is 5.35 m long and 2.53 m wide, with a total effective area of about 10 m². On the front side of each array, 9.5 m² of surface is covered with 3744 individual photovoltaic cells, which as an array produce 32 V. The solar arrays are capable of producing over 6000 W at Earth and around 2000 W at Mars (where the sunshine is weaker) at the start of the Primary Science Phase (PSP). At the end of the PSP, and at the Mars aphelion, the solar arrays are capable of producing 600 W. The latter value still allows the simultaneous operation of spacecraft (S/C) instruments and other subsystems [21, 26, 39]. Other solar array specs are not publicly available, however, given the technological heritage at the time of development and that the solar cells are triple junction with $\epsilon_{SA} = 0.26$, they are likely GaInP/GaAs/Ge cells [6]. However, a 6% reduction of the efficiency is estimated by the end of the mission due to degradation [36]. The solar panels have a composite structure, composed of the

materials shown in Figure 3. The inner and outer parts of the solar panel, alternatively referred to as *inboard* and *outboard* are made of different thicknesses of said materials, due to their differing thermal and structural requirements [14].

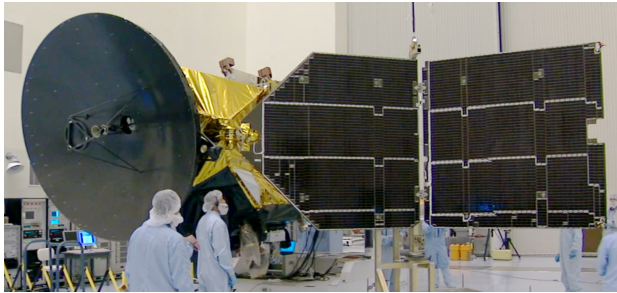


Figure 2: MRO's left solar array, semi-folded during a deployment test [29].

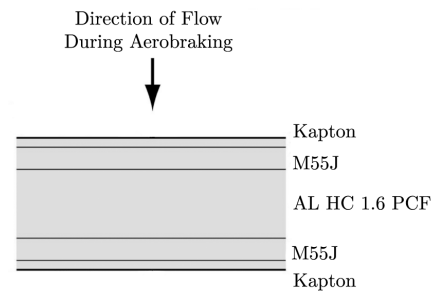


Figure 3: Cross section of the solar panel structure, excluding solar cells and connections [14].

The two-axis gimbal mechanisms of the solar arrays provide increased Sun-facing capabilities (as shown in Figure 4), reducing the average solar incidence angle drastically throughout the mission. While the solar arrays are designed to always point towards the Sun, there are cases wherein this is not possible and thus might pose an undervoltage risk. For example, on November 13th, 2017, during an Orbit Trim Maneuver (OTM), the solar arrays were pointed away from the Sun during a sunlit period and MRO subsequently entered eclipse, thereafter battery voltages dropped below critical levels, causing a full safing of the spacecraft [17]. An additional risk caused by this added articulation is collision of the solar arrays with High-Gain Antenna (HGA) or with the body of the MRO itself, the latter of which took place in the mission due to a mismanagement of the operationally-defined *keep-out zones* meant to prevent self-collision [24].

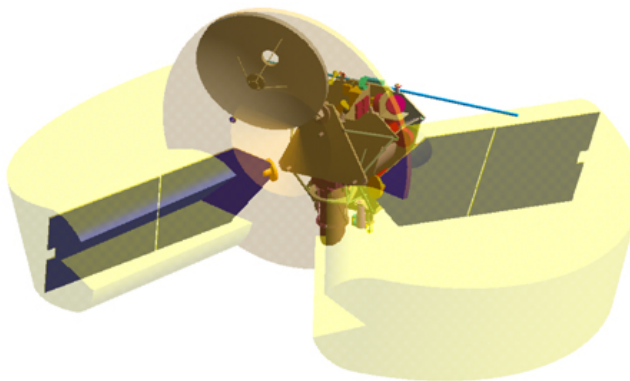


Figure 4: Geometry of MRO two-axis gimbal range of motion [24].

1.1.2 Batteries

MRO's orbit around Mars results in regular transitions between sunlight and shadow, necessitating a power system capable of enduring numerous charge-discharge cycles. The selection and design of batteries are driven by two important aspects, assuring the S/C survivability during Mars eclipses and hold out a high number of cycles, during the primary science and relay phase. The main power batteries of MRO are two Eagle-Picher 50 Ampere-hour, Common Pressure Vessel (CPV) Ni-H₂ batteries, composed of 11 cells each [33]. Each battery is capable of providing 1600 W for one hour at 32 V, even though the nominal voltage is typically set at 28 V. If the voltage drops below 20 V, the on board computer is not able to work and the spacecraft switches to safe mode [26]. The choice of

the Ni-H₂ batteries is driven by their long heritage in the space industry, offering reliability and high performances. This technology had accumulated over 30 years of research at the time, having amassed experience in the manufacturing of hydrogen electrode, separator material, electrolyte, overall battery design and life cycle tests [13]. Furthermore, these batteries offer high resistance against radiation and a wide temperature range of operability. The orbit of MRO with a period of around 2 hours results in many cycles over the mission lifetime, necessitating a low depth of discharge (DoD) for the batteries.

After MRO experienced several safings due to low voltage, moving the nominal orbit closer to the terminator (i.e. a later LST) was considered, however, this was evaluated to yield unsatisfactory science imaging results [17]. The MRO team has instead opted to initiate a process of battery reconditioning, wherein the battery and instrument parameters relating to power usage have been tweaked multiple times. Given the extended lifetime of the mission (as of the time of this work, MRO has completed over eighty-nine thousand orbits [28]), it is likely that the DoD has been reduced throughout the various battery management updates, such as the aforementioned battery reconditioning campaign, completed on June 10th, 2019 [17].

1.1.3 Power Control and Distribution Unit

The EPS includes a set of components dedicated to power distribution and control. The regulation is needed because of different aspects, such as load requirements, battery charge control, component degradation, and operational conditions diversity. There are no public sources dedicated to the harness and control logic of this subsystem, however, some considerations can be made based on typical standardized recommendations and previous missions. It is expected to have a centralized power distribution, with cables, fault protection and command decoders. The spacecraft bus voltage is assumed to be regulated at 28 V, consistent with NASA standard electrical system requirements. The EPS can be expected to operate on a Peak Power Tracking (PPT) logic, as it allows for more precise voltage control, which is critical due to the fluctuations in generated power throughout the sunlight-eclipse cycles [6, 11, 36].

1.2 Power Demands Across Mission Phases

Subsystem	Power Budget [W]						
	LEOP	Cruise	MOI	ATP	PSP (Sunlight)	(Eclipse)	Safe
TTMTC [35]	135	145.5	155	155	242	71	66
Propulsion [1, 15, 16]	-	122.27	122.27	62.87	62.87	62.87	13.72
AOCS [2, 3, 30, 31, 35]	58.32	63.92	30.92	30.92	120.92	63.6	40.32
EPS [11]	63.5	198.5	198.5	198.5	198.5	300	198.5
TCS [27]	30	250	50	280	30	300	30
OBDH [19]	10	10	10	10	10	10	10
Payload [4, 5, 7, 20, 39]	0	16	0	34	229.4	34	0
Total Power	296.8	790.2	566.7	737.3	893.7	841.5	358.5
Total (w/ margins)	356.2	948.2	680.0	884.7	1072.4	1009.8	430.2

Table 1: Subsystem power demand by mission phase for Mars Reconnaissance Orbiter.

Table 1 reports the average power consumption of the main subsystems across different mission phases or operational modes. It is evident that the power demand varies significantly depending on the spacecraft's operating conditions and requirements. Specifically, the **Tracking, Telemetry and Telecommand Subsystem (TTMTC)** estimation is based on literature sources, with certain assumptions applied. During the cruise phase, for example, the required power fluctuates considerably

due to instrumentation testing, trajectory correction maneuvers, and the increasing Earth-spacecraft distance. Therefore, an average between the minimum and maximum demand was considered. The HGA input power was estimated close to its maximum during the PSP phase, due to the long downlink distance and high data volume, while a lower value was assumed for the earlier phases. The power breakdown of each TTMTTC component is provided in the dedicated report. The **Attitude and Orbit Control Subsystem (AOCS)** demand generally remains well below 100 W, except during PSP, due to the high pointing accuracy and frequent attitude maneuvers required for target observations. During Mars Orbit Insertion (MOI) and Aerobraking and Transition Phase (ATP), power consumption is low, as thrusters are used instead of reaction wheels to provide the faster control response needed in those phases. In safe mode, the AOCS relies solely on the MIMU, sun sensors and low reaction wheel control torque. The values of **Thermal Control Subsystem (TCS)** are derived from data provided by NASA where a maximum heating power of 300 W is reported [27]. Then, considering on a case-by-case basis the actual heat loads from the various thermal sources, the corresponding electrical power requirements for each mission phase or operational mode are qualitatively assessed. In all phases a minimum of 30 W is imposed to consider the constant heating of the fuel pipes and tanks. For the **Payload** power, all testing and calibration activities are neglected due to their short duration. Therefore, non-zero power values appear only in PSP (full activation of payloads), during eclipse (stand-by mode of instruments), in cruise (ONC) and during ATP (instruments are switched on and in stand-by). **Propulsion** system is characterized by three different types of thrusters. Each of them is used for particular maneuvers during the mission, so different power requirements are demanded in each phase, according to values reported in datasheets [1, 15, 16]. The **On Board Data Handling (OBDH)** subsystem's power requirement is assumed to be equal to its On Board Computer (OBC), as the solid state recorder's power requirement is assumed to be low and is not publicly disclosed [19, 22]. The electric power profile of the control units of the **EPS** subsystem are not available, so the power values required by the battery control unit and that of the solar arrays, are estimated based on information obtained from other missions by the literature [11], therefore the power budget might be oversized with respect to reality.

1.3 Reverse Sizing

While MRO has far surpassed its projected lifetime, it was originally designed to carry out its primary mission until December 31, 2010, or until about 5.4 years after its launch [12].

Using MRO's SPICE data [28] and General Mission Analysis Tool (GMAT), the average eclipse duration (T_e , including penumbrae) over 1000 eclipse events is found to be around 37.1 minutes (± 47 seconds), whereas the average orbit time over three months is 111.9 minutes, leading to an average sunlit duration (T_L) per orbit of 74.8 minutes.

1.3.1 Solar Array Sizing

EoL energy capacity [Wh]	Mass [kg]	Volume [m ³]	Area [m ²]	Cell count	Series $\times$ Parallel
2174.21	96.49	1.1317	21.09	7645	13 $\times$ 590

Table 2: Solar array sizing results

The specific power of GaInP/GaAs/Ge cells is stated as 80 W/kg and their efficiency as $\epsilon_{SA} = 0.26$ [6]. Power at the beginning of life, P_{BoL} , is calculated using the solar irradiance at Mars orbit, as launch and cruise phases constitute a minority of the total lifetime. Line efficiencies corresponding to sunlit and eclipsed durations are $X_S = 0.80$ and $X_e = 0.60$ respectively [6]. Yearly degradation factor and inherent degradation factors are assumed $dp_y = 0.03$ and $I_d = 0.8$ respectively [6]. Given the

large range of movement and the two-axis freedom of motion of the solar wings, it is assumed that the average solar incidence angle over an orbit is 2° . An individual cell is assumed a fairly standard size of 27.5 cm^2 . From literature, the end-of-life (EoL) single cell voltage is around 2.3 V for similarly sized 3J solar cells [10, 6]. The solar irradiance on Mars is $P_{0, \text{Mars}} = 586.2 \text{ W/m}^2$ [25]. The design lifetime of around 5.4 years is considered. The mass per unit is obtained using the known thicknesses and the densities of the aforementioned constituent materials of the solar panels, in addition to the masses of individual cells (assumed 85 mg). As mentioned before, the inboard has significantly higher mass per unit than the outboard, at $\sigma_{\text{inboard}} = 2.83 \text{ kg/m}^2$ and $\sigma_{\text{outboard}} = 1.71 \text{ kg/m}^2$ respectively, without the solar cells. While exact dimensions are not available, and the outboards are slightly larger due to their lack of tapered corners, for the sake of simplicity, it is assumed that the inboard and the outboard are approximately of the same area and house an equal number of cells. As such, the total mass is:

$$M_{\text{SA}} = 2(\sigma_{\text{inboard}} \times A_{\text{inboard}}) + 2(\sigma_{\text{outboard}} \times A_{\text{outboard}}) + M_{\text{cells}} \quad (1)$$

Using the same thickness values that were used for the mass calculation, with the addition of the assumed cell thickness of $150 \mu\text{m}$, the volume of the arrays are obtained.

1.3.2 Battery Sizing

Batteries are the secondary source of power on MRO. The S/C is supplied with two Ni-H₂ battery packs. Each of them is composed of 11 cells in series [33]. Each cell has a voltage of 2.5 V in Direct Current (DC) [34]. The reported DoD is around 40% [26], which is confirmed by the graphical relations in the literature considering the design lifetime of the mission [37]. EoL battery efficiency η and packing efficiency loss factor μ are assumed to be 0.75 and 0.8 respectively, as typical values. Typical values of specific energy and energy density for Ni-H₂ batteries are respectively 55 W h/kg [38] and 50 Wh/l [6]. Once all the properties of the battery packs have been defined, knowing the eclipse duration and having obtained the required power in that period from Table 1, the main results of the sizing procedure, summarized in Table 3, can be computed. The computed results below are relative to a single pack of batteries, and a 20% margin is included on mass and volume:

Energy capacity [Wh]	Electric charge [Ah]	Mass [kg]	Volume [m ³]
1848	67.2	40.32	0.0444

Table 3: Single Battery Pack Sizing Results

1.3.3 Results Analysis

The cell count for the solar arrays is overestimated by a factor of 2.4%. However, changing the efficiency ε_{SA} value from 0.26 to 0.266 yields an exact match with the real cell count and a close agreement with the real populated area. The latter value is in fact slightly closer in line with the high-performing solar cells of the time [18]. This hints at the possibility that the reported solar cell efficiency is that of the EoL. Secondly, by visually inspecting images of the solar arrays, it can be seen that they are made up of strings of 18 cells, likely in series. Using the reported solar array supply voltage of 32 V [26] instead of the bus voltage of 28 V , the series voltage of 18×2.3 is still in excess. This may have been a measure against the voltage variations of the cells with temperature [23], combined with the additional voltage that would be required to account for inherent degradation at EoL. Additionally, if a maneuver needs to be executed during a satellite day (particularly in PSP), the battery recharge time will be drastically reduced, requiring higher voltages to compensate. The calculated mass of a singular solar array is 48.25 kg , closely in agreement with the real mass of 48.3 kg [8].

Regarding the battery sizing, the results obtained can be considered reasonable in terms of both mass and volume. The only results that can be directly compared with real data from the literature are the number of cells in series and the battery charge. The number of cells in series computed during the sizing matches the actual value, which suggests that the input data used in the sizing process—such as the total system voltage and the single cell voltage—are likely accurate [33]. On the other hand, the battery pack charge obtained from the sizing is higher than the value reported in the literature [33]. This difference is linked to an overestimation of the energy capacity, which is likely due to inaccuracies in defining the battery efficiency and packing efficiency loss factor values. Nevertheless, considering that this is a preliminary analysis, the battery sizing can be considered sufficiently comprehensive.

1.3.4 Mass, Volume, Power and Data Budget

The mass and volume budget are reported below:

Component	Mass [kg]	Volume [m ³]
Solar arrays	96.49 (2×48.25)	1.1317 (2×0.5658)
Batteries	80.64 (2×40.32)	0.0889 (2×0.0444)
Total	177.13	1.2206
Total (w/ 20% margins)	212.56	1.4647
Total margined with cables	255.07	1.7576

Table 4: EPS mass and volume budget.

The most common power demand from the EPS is estimated at 198.5 W, while the peak power demand is around 300 W which only occurs during eclipse [11]. The MRO query servers operate at a sampling rate of 0.2 Hz for delta - Differential One-way Range (Δ DOR) measurements [32]. Total data rate can be estimated assuming the same sampling rate for telemetry monitoring, with telemetry data containing 10 parameters (such as battery output voltage, solar array output voltage, main bus voltage etc.) averaging roughly 2 bytes each, as a mix of binary and single-precision float values. The resulting estimate is 345.6 kilobytes per day.

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Acronyms

ΔDOR delta - Differential One-way Range. 6

AOCS Attitude and Orbit Control Subsystem. 3, 4

ATP Aerobraking and Transition Phase. 3, 4

CPV Common Pressure Vessel. 2

DC Direct Current. 5

DoD depth of discharge. 3, 5

EoL end-of-life. 4, 5

EPS Electric Power Subsystem. 1, 3, 4, 6

HGA High-Gain Antenna. 2, 4

LEOP Launch and Early Orbit Phase. 3

LST Local Solar Time. 3

MIMU Miniature Inertial Measurement Unit. 4

MOI Mars Orbit Insertion. 3, 4

MRO Mars Reconnaissance Orbiter. 1–6

OBC On Board Computer. 4

OBDH On Board Data Handling. 3, 4

OTM Orbit Trim Maneuver. 2

PPT Peak Power Tracking. 3

PSP Primary Science Phase. 1, 3–5

S/C spacecraft. 1, 2, 5

TCS Thermal Control Subsystem. 3, 4

TTMTC Tracking, Telemetry and Telecommand Subsystem. 3, 4